

Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A) Mech/ Civil (NEP) Sem - II
Examination June/July 2025
Basics of Electrical and Electronics Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any four questions from Part B: Q.2 to Q.7.
- 3) Use of non-programmable calculator is allowed

PART-A**Q1 Solve the following questions****20**

- a) State Ohm's law along with its formula.
- b) Define Electric potential and Resistance along with its formula.
- c) Fleming's Right-Hand Rule is used for:
 - i. Motors
 - ii. Transformers
 - iii. Induced EMF
 - iv. Capacitance
- d) In a magnetic circuit, reluctance is analogous to:
 - i. Voltage
 - ii. Resistance
 - iii. Current
 - iv. Inductance
- e) Which part of a transformer is used to provide insulation and cooling?
 - i. Core
 - ii. Laminations
 - iii. Oil
 - iv. Yoke
- f) The stator of an induction motor is:
 - i. Stationary
 - ii. Rotating
 - iii. Insulated
 - iv. Intermittent
- g) Which of the following is an active component?
 - i. Resistor
 - ii. Capacitor
 - iii. Transistor
 - iv. Inductor
- h) The Zener diode operates in:
 - i. Forward bias
 - ii. Reverse bias
 - iii. Both
 - iv. None
- i) Binary code for decimal no 14 is
 - i. 1110
 - ii. 0101
 - iii. 1000
 - iv. None of the above
- j) Which logic gate has the output true only when both inputs are true?
 - i. OR
 - ii. AND
 - iii. EX-OR
 - iv. NOT

PART-B

- Q2** a) What is Electric Potential and Potential Difference? Explain how they relate to Work and Energy in an electric circuit. **05**
 b) Define the following terms with one example each: (i) Cycle, (ii) Time period, (iii) Frequency, (iv) Amplitude, (v) Phase difference. **05**
- Q3** a) Compare electrical circuit and magnetic circuits. **05**
 b) State and explain Fleming's Left-Hand Rule. How is it used to determine the direction of force in an electric motor? **05**
- Q4** a) Explain the construction and working of an alternator. What are its main components and how is EMF generated? **05**
 b) Classify transformers based on construction and application. Give one example of each type. **05**
- Q5** a) Explain the construction and working of Bipolar Junction Transistor (BJT). Mention its types and configurations. **05**
 b) Compare PN junction diode and Zener diode in terms of symbol, characteristics, and applications. **05**
- Q6** a) Explain Number systems in detail along with its Examples. **05**
 b) Differentiate between combinational circuits and sequential circuits **05**
- Q7** a) Write short note on Temperature measurement transducers. **05**
 b) Illustrate LVDT in detail. **05**